Algorithms for Data Science
Cheatsheet for Exercises 6

Notations in Graph Theory:

- n is the number of vertices.
- m is the number of edges.
- If you draw a graph, you are allowed to draw edges also as curves (if not otherwise noted).
- *node* is a different word for *vertex*.

- A *path* is a sequence of consecutive edges, where the next edge starts where the last edge ends.
  - The *length* of a path is the number of its edges.
  - The *cost* (*weight*) of a path is the total weight of its edges.
  - A path is *simple* if it has no repeating vertices.
- A *cycle* is a simple path that ends where it starts.
- A *tour* is a (not necessarily simple) path that ends where it starts and that visits all the vertices of the graph.
- A graph is *connected* if one can reach any vertex from any other one by a path.
- A *tree* is a connected undirected graph without cycles.

- A *subgraph* is any graph that contains a subset of the vertices and edges of the original graph. For example, every graph is a subgraph of itself.
- A *spanning subgraph* of a connected graph is any connected subgraph containing all the vertices.
- A spanning subgraph is *minimal* if its total edge weight is minimal.

If not otherwise noted, we assume graphs to be undirected and unweighted and to have neither parallel edges nor self-loops (a self-loop is an edge that starts and ends in the same vertex).